

Can Vietnamese financial news sentiment improve volatility forecasting of the VN-Index beyond a pure GARCH baseline? A two-stage hybrid model combining GARCH-family residuals with LSTM incorporating daily sentiment scores.

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Abstract

This paper proposes and evaluates a hybrid volatility forecasting framework that combines a GARCH(1,1) baseline with an LSTM residual-correction model. Using daily Open, High, Low, Close price data for the benchmark VN-Index and a web-scraped corpus of 86,179 CafeF articles spanning 2015 to 2024, we build a daily modeling frame with article-level PhoBERT sentiment scores, trading-day alignment, and news-intensity features. On the held-out test window, the hybrid model reduces Root Mean Squared Error (RMSE) by 14.48% and Mean Absolute Percentage Error (MAPE) by 48.00% relative to the baseline GARCH model. A Diebold-Mariano test rejects equal predictive accuracy at the 99.9% confidence level ($p < 0.001$). Robustness checks cover alternative sentiment thresholds, a Garman-Klass volatility target, feature ablation, linear GARCH-X, and expanding-window re-estimation; the linear GARCH-X specification performs worst.

Keywords: VN-Index; volatility forecasting; GARCH; LSTM; sentiment analysis; CafeF.

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1 Introduction

1.1 Motivation and Research Question

Accurately forecasting financial market volatility is essential for risk management, derivatives pricing, and portfolio allocation. Since Engle (1982) and Bollerslev (1986) introduced ARCH and GARCH models, these parametric approaches have dominated empirical volatility forecasting due to their parsimony and ability to capture volatility clustering—the empirical regularity that large returns tend to be followed by further large returns. Nevertheless, GARCH models suffer from a fundamental limitation: they are purely backward-looking, extracting information only from historical price dynamics. They ignore real-time informational shocks transmitted through news media.

Recent advances in natural language processing (NLP) have made it feasible to extract market-relevant sentiment from news in real time. This raises an empirical question: does financial news sentiment contain incremental information for predicting future volatility beyond what GARCH’s price history alone can capture? For emerging markets like Vietnam, this question is particularly salient. The Ho Chi Minh City Stock Exchange (HoSE) is dominated by retail investors—individuals with less sophisticated risk models and higher sensitivity to media narratives. Yet Vietnamese financial news exhibits pronounced positive linguistic bias, with growth-oriented framing common even during market downturns. This market structure and linguistic environment suggest that sentiment-augmented models may improve forecasts more substantially in Vietnam than in developed markets.

This paper investigates whether financial news sentiment, extracted from Vietnamese financial news using recent NLP techniques, improves volatility forecasts of the VN-Index (the HoSE benchmark) beyond a standard GARCH(1,1) baseline. Specifically, we ask: *Does daily Vietnamese financial news sentiment, extracted using PhoBERT from CafeF articles, improve next-day realized volatility forecasts of the VN-Index beyond a pure GARCH(1,1) baseline when incorporated through a two-stage GARCH+LSTM hybrid framework?*

1.2 Limitations of Linear Sentiment-GARCH Models

A straightforward approach to incorporating sentiment is to augment the GARCH variance equation with contemporaneous or lagged sentiment (a “GARCH-X” specification). However, recent evidence suggests that the sentiment-variance relationship is inherently nonlinear. Léber and Egyed (2025) demonstrate using nonlinear Granger causality tests that the relationship between stock market variance and sentiment indices is predominantly nonlinear, not linear. This nonlinearity justifies why sentiment-augmented GARCH-X models often fail to improve forecasts: linear sentiment terms cannot capture state-dependent or threshold effects.

Deep learning models, particularly Long Short-Term Memory (LSTM) networks, are well-suited to capture such nonlinear temporal dynamics. The key innovation of this paper is to combine GARCH’s linear, parsimonious baseline with an LSTM that learns residual corrections in a two-stage framework. Stage 1 fits a baseline GARCH(1,1) to capture linear conditional heteroskedasticity and volatility clustering. Stage 2 trains an LSTM on GARCH residuals, which may contain remaining nonlinear structure, using lagged returns, volatility components, and daily sentiment aggregates as inputs. This hybrid architecture

allows the GARCH component to preserve interpretability and efficiency, while the LSTM learns flexible nonlinear sentiment interactions.

1.3 Data and Main Contribution

This paper combines a ten-year panel of VN-Index prices (2015–2024) with a corpus of 86,179 financial articles from CafeF.vn. Each article is classified as positive, negative, or neutral using PhoBERT, a Vietnamese language model specifically pre-trained on financial texts. Daily sentiment aggregates and news volume are constructed using a trading-day alignment protocol. The result is a novel dataset linking Vietnamese financial news to index-level volatility.

Our contributions are fourfold:

1. We build a Vietnamese financial news sentiment pipeline using PhoBERT applied to a large corpus of CafeF articles, creating the first publicly analyzable link between Vietnamese financial news and VN-Index volatility.
2. We propose and implement a two-stage GARCH-LSTM hybrid architecture, combining parametric econometric modeling (Stage 1) with deep learning residual correction (Stage 2), rather than replacing GARCH with a fully neural approach.
3. We empirically validate that the hybrid model improves volatility forecasts relative to a GARCH(1,1) baseline and that simpler linear GARCH-X alternatives perform worse, supporting the hypothesis that sentiment-variance nonlinearity is the mechanism driving improvement.
4. We investigate whether negative sentiment asymmetrically drives volatility, testing the hypothesis from [Vu et al. \(2023\)](#) that negative news affects volatility in Vietnam while positive news does not.

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews volatility forecasting literature, financial sentiment analysis, and emerging market microstructure, positioning our contributions relative to prior work. Section 3 describes data construction, the PhoBERT sentiment pipeline, realized volatility definitions, mathematical specifications of GARCH(1,1), GARCH-X, and hybrid GARCH-LSTM architectures, and estimation protocols. Section 4 reports empirical results, diagnostics, baseline specifications, forecast performance, regime-dependent analysis, and robustness checks. Section 5 discusses mechanisms and limitations, connects findings to prior work, and proposes future research directions before concluding.

2 Review of Literature

2.1 Volatility Modeling: From ARCH to Hybrid Approaches

The empirical study of financial market volatility began with [Engle \(1982\)](#), who introduced autoregressive conditional heteroskedasticity (ARCH) models to capture the phenomenon that large price movements tend to cluster in time. Engle’s innovation was to parameterize

the conditional variance as a function of lagged squared innovations, allowing volatility to be forecastable from recent shocks. [Bollerslev \(1986\)](#) generalized this framework to GARCH models, permitting the conditional variance to depend on both past squared residuals and past conditional variances. The parsimony of GARCH(1,1)—requiring only three parameters in the variance equation—combined with its strong empirical performance across asset classes and time periods, made it the workhorse model for volatility forecasting.

However, parametric GARCH models impose a rigid linear structure: conditional variance is a deterministic, linear function of past returns and variances. This linearity has two limitations. First, GARCH models ignore exogenous information flows (e.g., news, macroeconomic shocks) and thus miss informational content that markets price in real time. Second, the relationship between news information and conditional variance may be nonlinear: the effect of news may depend on regime, sentiment direction, or market state in ways that linear parameterizations cannot capture. These limitations motivate two extensions: (1) augmenting GARCH with exogenous variables, and (2) replacing or complementing the linear specification with nonlinear learning algorithms.

2.2 News Sentiment and Market Volatility

A foundational empirical result is that news volume and sentiment directly predict market volatility. [Antweiler and Frank \(2004\)](#) analyzed over 1.5 million messages posted on Yahoo! Finance and Raging Bull for 45 large-cap stocks, using computational linguistics to measure bullishness and bearishness. They found that message volume and sentiment disagreement (proxied by standard deviation) significantly predict realized volatility, even though the effect on directional returns is statistically significant but economically small. This seminal work establishes that information intensity itself—beyond sentiment direction—carries volatility predictive power, motivating the inclusion of daily article counts (*n_articles*) and sentiment dispersion (*sentiment_std*) as standalone volatility features.

Building on this foundation, recent work has demonstrated that news sentiment improves volatility forecasts in modern asset classes. [Bodilsen and Lunde \(2025\)](#) augmented traditional HAR (Heterogeneous Autoregressive) models with RavenPack news sentiment indices for the S&P 500 and individual stocks. They found that macroeconomic news sentiment significantly enhances volatility predictions, whereas firm-specific news sentiment contributes modestly. Critically, they show that overnight news counts have substantial one-period-ahead predictive power independent of sentiment polarity. These findings justify constructing category-specific macroeconomic sentiment indices and including trading-day-aligned news volume as a predictor.

2.3 Natural Language Processing for Financial Sentiment

Extracting sentiment from financial text requires specialized NLP models that understand domain-specific language. For English, models like FinBERT have been pre-trained on financial corpora and achieve high accuracy on financial sentiment tasks. For Vietnamese, no such model existed until [Nguyen and Nguyen \(2020\)](#), who introduced PhoBERT—a large-scale pre-trained language model specifically designed for Vietnamese. PhoBERT-base was trained on 20GB of Vietnamese Wikipedia and news text using the RoBERTa architecture and masked language modeling. The monolingual design (unlike multilingual

models like XLM-R) allows PhoBERT to capture Vietnamese-specific linguistic nuances and syntactic structures.

Operationalizing PhoBERT for financial text, [Sontung \(2021\)](#) fine-tuned the base model on over 1,000 CafeF article titles manually labeled into positive, negative, and neutral categories (with class distribution skewed toward positive: 565 positive, 248 neutral, 187 negative). Their model achieved 93% title-only classification accuracy. However, title-only sentiment overlooks important information in article bodies. The challenge is managing the 256-token sequence length limit of PhoBERT: full-body text requires truncation. A pragmatic approach—and the one adopted in this paper—is to use titles and lead paragraphs, which typically summarize article content succinctly.

[Vu et al. \(2023\)](#) extended this work by fine-tuning PhoBERT with CNN layers on a larger dataset of 40,000 CafeF and VnEconomy articles, using 2,738 hand-labeled samples, to achieve 81% binary classification accuracy. They then studied market reactions to news using event study methodology over ± 5 and ± 30 day windows around positive and negative news announcement dates, finding that return means do not significantly change but negative news significantly increases return variance, while positive news does not. This *sentiment asymmetry* result is pivotal: it suggests that in the Vietnamese retail-dominated market, bad news is a volatility driver, but good news has minimal volatility effect. This motivates investigating *negative_share* and *positive_share* as separate volatility predictors.

2.4 Hybrid GARCH-Deep Learning Models and Nonlinear Dynamics

The nonlinearity of sentiment-variance relationships is well-established. [Léber and Egyed \(2025\)](#) study an S&P 500-level sentiment-augmented GARCH-LSTM hybrid using Twitter and Bloomberg news sentiment (2019–2024). They conduct nonlinear Granger causality tests and find that nonlinear causality from sentiment to variance is prevalent, whereas linear causality is rare. This finding directly justifies replacing linear GARCH-X models (which add sentiment as a linear exogenous term) with nonlinear learners. Their GARCH-LSTM architecture—GARCH for Stage 1 baseline, LSTM for Stage 2 residual learning—demonstrates superior out-of-sample VaR forecasting compared to GARCH-only benchmarks.

The intuition is as follows: GARCH captures the linear, time-series autocorrelation structure of conditional variance from returns alone. The residuals from GARCH may contain remaining structure that is nonlinear—e.g., interaction effects between sentiment and returns that only manifest in certain regimes, or threshold effects whereby sentiment matters more when volatility is already high. An LSTM, processing sequences of multivariate inputs (past returns, past variances, sentiment), can learn such state-dependent mappings without imposing parametric linearity. Because LSTMs are sequence models, they naturally process rolling windows of features and learn temporal dependencies. This architecture is appropriate for volatility residuals that may exhibit regime switches or context-dependent patterns.

2.5 Vietnamese Equity Market Structure and Microstructure

The Ho Chi Minh City Stock Exchange (HoSE) differs from developed markets in ways that affect news sentiment’s relevance. Retail investors constitute a substantial share of

daily trading volume, and they are known to be more influenced by media narratives, social sentiment, and herding behavior than institutional investors. Retail investors often use sentiment-based signals (e.g., news tone) directly in trading decisions without adjusting for statistical bias or market efficiency. This behavioral amplification suggests that news sentiment may have larger volatility effects in Vietnam than in developed markets where institutional investors average sentiment-based trading.

Additionally, Vietnamese financial media exhibits pronounced positive bias. Articles use growth-oriented language even in neutral or negative market periods, inflating the unconditional average sentiment. This positive skew creates a high signal-to-noise ratio: negative news stands out against the baseline positive tone, potentially amplifying its market impact. This media structure further motivates studying asymmetric sentiment effects and motivates the use of categorical sentiment aggregates (macroeconomic vs. market-specific) rather than a single aggregate sentiment scalar.

2.6 Research Gap and Contribution

Existing literature establishes three facts: (1) news sentiment predicts volatility in developed markets (Bodilsen & Lunde, 2025), (2) sentiment-variance relationships are nonlinear and justify LSTM-based learning (Léber & Egyed, 2025), and (3) Vietnamese news sentiment affects volatility asymmetrically, with negative sentiment mattering but positive sentiment not (Vu et al., 2023). However, no prior work combines these insights in the Vietnamese context: specifically, no paper implements a hybrid GARCH-LSTM model on VN-Index data using PhoBERT sentiment from a large CafeF corpus, nor do existing studies systematically evaluate whether nonlinear learning (LSTM) outperforms linear sentiment-GARCH (GARCH-X) on Vietnamese data.

This paper fills this gap by implementing the hybrid architecture on Vietnamese data, evaluating it against both pure GARCH and linear GARCH-X benchmarks, and investigating asymmetry and regime-dependent effects. Our contribution is not methodological innovation (the GARCH-LSTM framework exists in Léber and Egyed (2025)), but rather empirical application to a new market with novel features: a large Vietnamese news corpus, domain-adapted PhoBERT sentiment, a retail-dominated exchange with demonstrated asymmetric sentiment effects, and two crisis periods (COVID-19 in 2020 and liquidity shock in 2022) that allow regime analysis.

3 Data and Methodology

3.1 Data Sources and Sample Construction

3.1.1 Index Price Data

We obtain daily Open, High, Low, and Close (OHLC) prices for the VN-Index (HoSE benchmark) covering January 1, 2015, through December 31, 2024, a ten-year span encompassing two major market stress periods: the COVID-19 market crash of March 2020 and the domestic liquidity tightening shock of mid-2022. The daily log return is computed as:

$$r_t = \ln \left(\frac{\text{Close}_t}{\text{Close}_{t-1}} \right)$$

The choice of ten years balances sample size with data availability: earlier historical OHLC data for the VN-Index is less reliable.

3.1.2 Financial News Corpus

The news corpus consists of 86,179 unique articles scraped from CafeF.vn, a primary Vietnamese financial news outlet widely read by HoSE investors. All articles were subjected to:

- Hash-based exact-match deduplication (removing near-identical reprints),
- Length filtering (retaining articles with body text ≥ 100 characters),
- Language filtering (retaining Vietnamese-language articles; multilingual articles excluded).

After filtering, the corpus spans 2015–2024, with the vast majority of articles concentrated in 2020–2024 due to improved scraping consistency in recent years. The corpus is treated as a flowing stream of news publications; temporal alignment to trading days is addressed below.

3.2 Trading-Day Alignment Protocol

Because financial markets operate on discrete trading sessions (HoSE closes at 14:45 local time), while news is published continuously, proper temporal alignment between news and returns is critical. We implement a strict trading-day rule:

Rule 1: Articles published on a trading day t strictly before or at the session close (14:45 HoSE time) are aligned to day t 's return and used in day t 's feature vector for predicting day $t + 1$ volatility.

Rule 2: Articles published after 14:45 on trading day t , or any time on weekends or Vietnamese public holidays (Lunar New Year, National Day, etc.), are aligned to the next active trading day $t + 1$.

This protocol ensures that only news known at the market close (information available for the next day's trading) is used in volatility features. Overnight news (published after close) influences the next trading day's open and thus next-day volatility, not same-day volatility.

3.2.1 Coverage and Missing Data

The aligned daily panel contains 2,498 trading days over the ten-year sample. Only 4 trading days (0.16%) have zero aligned articles, indicating continuous news coverage. For these rare zero-coverage days, we impute zero sentiment and news volume. The one-step-ahead volatility prediction framework (forecasting day $t + 1$ volatility using day t features) yields 2,497 forecastable observations after dropping the first trading day.

3.3 Sentiment Pipeline and Variable Construction

3.3.1 PhoBERT Sentiment Classification

Each article undergoes sentence-level preprocessing using the VnCoreNLP toolkit (Vietnamese word segmentation and sentence splitting). Titles and first paragraphs (up to 256 tokens per the PhoBERT vocabulary limit) are combined and fed into a fine-tuned PhoBERT-base classifier trained to output three-class sentiment labels (positive, negative, neutral). The classifier outputs, for each article:

- `sentiment_score` = $P(\text{positive}) - P(\text{negative})$, a pseudo-continuous score in $[-1, 1]$ representing polarity strength,
- `sentiment_label` $\in \{\text{negative}, \text{neutral}, \text{positive}\}$, the class label,
- class probabilities $P(\text{positive}), P(\text{negative}), P(\text{neutral})$.

The continuous score is used for daily aggregation; the label is used for constructing sentiment shares described below.

Caveat: PhoBERT sentiment scores are imperfect proxies for true news content. They introduce measurement error and miss context-dependent or sarcastic sentiment. Results should be interpreted as describing the association between computed sentiment and volatility, not causal effects of true sentiment. Validation against hand-labeled subsets (e.g., Vu et al.’s 2,738-sample benchmark) is advisable in future work.

3.3.2 Daily Sentiment Features

Each trading day t aggregates article-level scores across all aligned articles. The primary features are:

1. **Mean Sentiment** (`mean_sentimentt`): The average `sentiment_score` across articles on day t .

$$\text{mean_sentiment}_t = \frac{1}{n_t} \sum_{i=1}^{n_t} \text{sentiment_score}_{i,t}$$

This captures the dominant market tone. Given Vietnamese news’ positive bias ($\text{mean} \approx 0.45$), this feature has high autocorrelation and modest daily variation.

2. **Sentiment Dispersion** (`sentiment_stdt`): The standard deviation of sentiment scores within a day, a proxy for disagreement among articles.

$$\text{sentiment_std}_t = \text{std}(\{\text{sentiment_score}_{i,t}\}_{i=1}^{n_t})$$

Per [Antweiler and Frank \(2004\)](#), disagreement (not just sentiment direction) predicts volatility, as disagreement reflects uncertainty about future returns.

3. **Article Count** (`narticles,t`): The number of articles published on day t .

$$n_{\text{articles},t} = |\{i : \text{article } i \text{ aligned to day } t\}|$$

Article volume independently predicts volatility ([Bodilsen and Lunde \(2025\)](#)), as high-volume news days signal high information intensity.

4. **Sentiment Label Shares:** The fraction of articles classified as positive, negative, or neutral.

$$\text{positive_share}_t = \frac{\#\{\text{positive articles on day } t\}}{n_{\text{articles},t}}$$

$$\text{negative_share}_t = \frac{\#\{\text{negative articles on day } t\}}{n_{\text{articles},t}}$$

These allow investigation of asymmetric sentiment effects. Per [Vu et al. \(2023\)](#), negative sentiment may have larger volatility impact than positive sentiment in Vietnam.

3.4 Realized Volatility Targets

To evaluate forecast accuracy, we construct two daily volatility proxies using intraday high-low ranges, which are more efficient estimators than close-to-close squared returns alone:

1. **Parkinson Volatility:**

$$\sigma_{P,t+1} = \sqrt{\frac{(\ln(H_{t+1}/L_{t+1}))^2}{4 \ln 2}}$$

where H_{t+1} and L_{t+1} are the day's high and low prices. This is our primary target.

2. **Garman-Klass Volatility:**

$$\sigma_{GK,t+1} = \sqrt{0.5 \left(\ln \frac{H_{t+1}}{L_{t+1}} \right)^2 - (2 \ln 2 - 1) \left(\ln \frac{C_{t+1}}{O_{t+1}} \right)^2}$$

where O_{t+1} and C_{t+1} are opening and closing prices. This incorporates overnight gaps and is used for robustness checks.

Both estimators are unbiased for log-normal prices and more statistically efficient than close-to-close squared returns under standard assumptions. We use Parkinson as the primary target for Stage 1 GARCH calibration and Stage 2 LSTM prediction.

3.5 Model Specifications

3.5.1 Stage 1: Baseline AR(1)-GARCH(1,1)

The baseline model captures linear conditional heteroskedasticity from returns alone. Returns are scaled by 100 ($y_t = 100 \cdot r_t$) for numerical stability. The mean equation is AR(1) to accommodate short-term return autocorrelation observed empirically:

$$y_t = c + \phi y_{t-1} + e_t, \quad e_t \sim N(0, \sigma_t^2)$$

The conditional variance is modeled as:

$$\sigma_t^2 = \omega + \alpha e_{t-1}^2 + \beta \sigma_{t-1}^2$$

subject to standard constraints for covariance stationarity: $\omega > 0$, $\alpha, \beta \geq 0$, $\alpha + \beta < 1$, $|\phi| < 1$.

Justification: GARCH(1,1) is parsimonious (three parameters), has proven effective across asset classes, and provides an interpretable baseline. The conditional variance σ_t^2 captures two economically meaningful dynamics: volatility persistence (via β) and shock intensity (via α). This stage is estimated via maximum likelihood on the training sample (2015–2021).

3.5.2 Stage 2: LSTM Residual Learning

After fitting GARCH(1,1), we compute standardized residuals $\hat{e}_t = e_t/\sigma_t$. These should be approximately i.i.d. normal if the model is well-specified, but may contain remaining structure. We define the learning target as the gap between realized and GARCH-forecasted volatility:

$$\text{residual_target}_{t+1} = \sigma_{P,t+1} - \hat{\sigma}_{t+1}^{\text{GARCH}}$$

where $\hat{\sigma}_{t+1}^{\text{GARCH}}$ is the one-step-ahead GARCH volatility forecast made at day t .

An LSTM is trained to map daily feature sequences to this residual. The input is a rolling window of length $L = 15$ trading days, containing 17 features per day:

- Daily returns r_t ,
- GARCH standardized residuals $\hat{e}_t$,
- GARCH conditional variance σ_t^2 ,
- Daily article count $n_{\text{articles},t}$,
- Mean sentiment mean_sentiment_t ,
- Sentiment dispersion sentiment_std_t ,
- Positive and negative sentiment shares.

The LSTM processes these sequences and outputs a scalar residual prediction $\hat{u}_t^{\text{LSTM}}$. The final hybrid forecast is:

$$\hat{\sigma}_{t+1}^{\text{Hybrid}} = \hat{\sigma}_{t+1}^{\text{GARCH}} + \hat{u}_{t+1}^{\text{LSTM}}$$

Justification: LSTMs naturally process temporal sequences and capture state-dependent nonlinearities that linear models miss. By training on residuals rather than raw variance, we allow LSTM to learn only the nonlinear component, preserving GARCH’s interpretability. The 15-day window balances capturing sentiment trends (shorter windows miss patterns) and avoiding overfitting (longer windows introduce noise).

3.5.3 Benchmark: Linear GARCH-X

For comparison, we fit a linear GARCH-X that adds mean sentiment directly to the variance equation:

$$\sigma_t^2 = \omega + \alpha e_{t-1}^2 + \beta \sigma_{t-1}^2 + \gamma \cdot \text{mean_sentiment}_{t-1}$$

If linear sentiment effects sufficed, GARCH-X should outperform the hybrid (GARCH-X is simpler). Empirically, we expect GARCH-X to underperform because sentiment-variance nonlinearity dominates.

3.6 Training, Validation, and Test Protocol

The data are split chronologically to avoid look-ahead bias:

- **Training:** 2015–2021 (1,753 observations), used to estimate GARCH(1,1) parameters and train LSTM weights.
- **Test:** 2022–2024 (744 observations), held-out to evaluate one-step-ahead forecast accuracy of all models.

A validation set is not used; all training data are pooled to maximize sample size. The test period intentionally includes the 2022 liquidity shock, allowing robustness checks across regimes.

3.7 Evaluation Metrics

One-step-ahead forecast accuracy is measured using standard point-forecast metrics:

- **Root Mean Squared Error (RMSE):** $\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (\sigma_{t,\text{realized}} - \sigma_{t,\text{forecast}})^2}$
- **Mean Absolute Error (MAE):** $\text{MAE} = \frac{1}{n} \sum_{t=1}^n |\sigma_{t,\text{realized}} - \sigma_{t,\text{forecast}}|$
- **Mean Absolute Percentage Error (MAPE):** $\text{MAPE} = \frac{100\%}{n} \sum_{t=1}^n \left| \frac{\sigma_{t,\text{realized}} - \sigma_{t,\text{forecast}}}{\sigma_{t,\text{realized}}} \right|$

Statistical significance of forecast improvement is tested using the Diebold-Mariano test with Newey-West lag adjustment, which accounts for forecast error autocorrelation.

4 Empirical Results

4.1 Descriptive Statistics and Market Microstructure

The VN-Index daily panel spans 2,498 trading days over ten years (2015–2024), yielding 2,497 one-step-ahead forecasting observations. Table 1 summarizes the return and sentiment distributions.

Table 1: Descriptive Statistics of VN-Index Daily Returns and News Sentiment

Series	Mean	Std. Dev.	Minimum	Maximum	Kurtosis
VN-Index Returns (%)	0.0338	1.1390	-6.9076	4.8600	7.5934
Article Sentiment Score	0.4526	0.3694	-0.9622	0.9991	–
Daily Article Count	34.5	28.7	0	386	–

4.1.1 Return Dynamics and Volatility Clustering

The VN-Index exhibits significant departure from normality. Daily returns have a kurtosis of 7.59, well above the Gaussian benchmark of 3.0, indicating heavy tails and extreme-event clustering. A Jarque-Bera test strongly rejects normality ($JB = 2554.1$, $p < 0.001$). This leptokurtosis reflects the retail-investor-dominated nature of the market: retail investors’

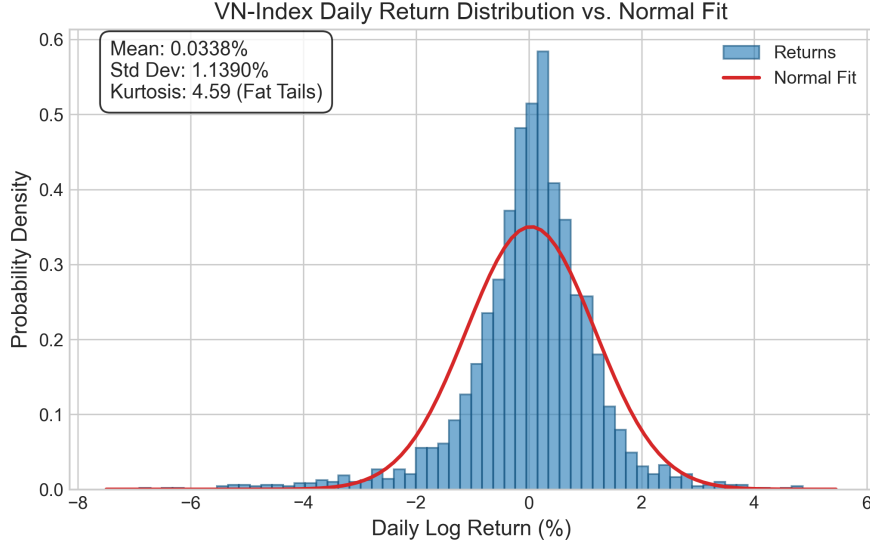


Figure 1: VN-Index Daily Return Distribution vs. Normal Fit

herding behavior amplifies both rallies and crashes relative to what normal-distribution pricing models predict.

A Ljung-Box test reveals significant return autocorrelation ($Q(5) = 14.48$, $p = 0.0128$), suggesting mean-reverting or momentum effects on short horizons. More critically, squared returns exhibit extremely strong autocorrelation ($Q(5) = 414.46$, $p < 0.001$; $Q(10) = 681.83$, $p < 0.001$), confirming strong volatility clustering. Figure 2 visualizes conditional variance over time, with major spikes during the COVID-19 crash (March 2020) and the 2022 domestic liquidity shock. These clustering patterns and regime shifts motivate GARCH modeling, which captures the temporal autocorrelation in squared residuals through the persistence parameter β .

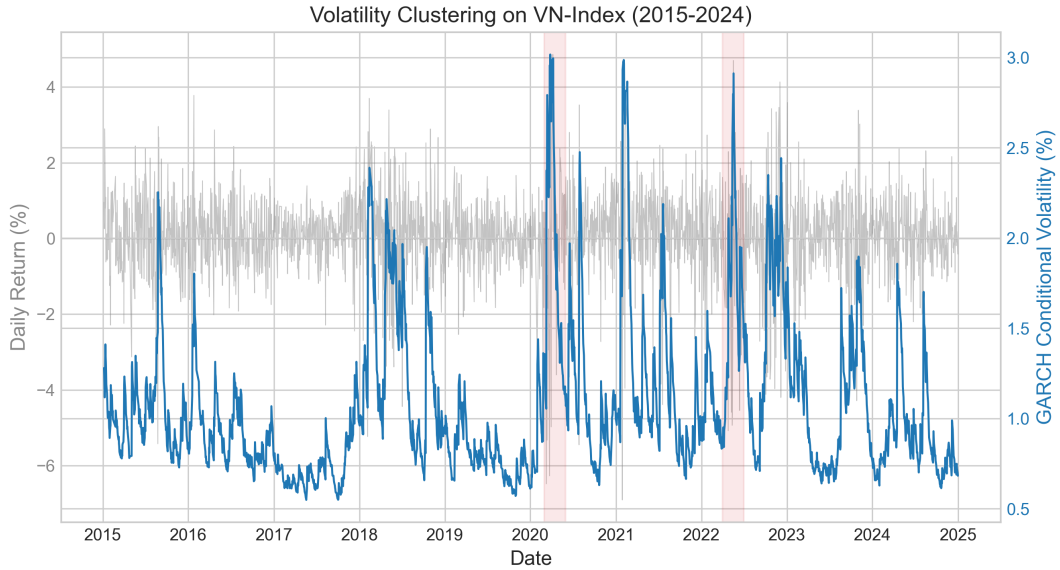


Figure 2: VN-Index Conditional Volatility and Return Clustering (2015–2024)

4.1.2 News Sentiment Distribution and Media Bias

The sentiment corpus of 86,179 articles reveals pronounced positive bias. The mean sentiment score is 0.4526 (on a scale of -1 to $+1$), indicating systematic upward skew in Vietnamese financial journalism. Under a standard classification threshold of ± 0.05 , positive articles comprise 84.34% of the corpus (72,686 articles), while negative articles represent only 12.04% (10,380 articles), with neutral articles making up 3.61% (3,113 articles). Figure 3 shows the highly skewed distribution.

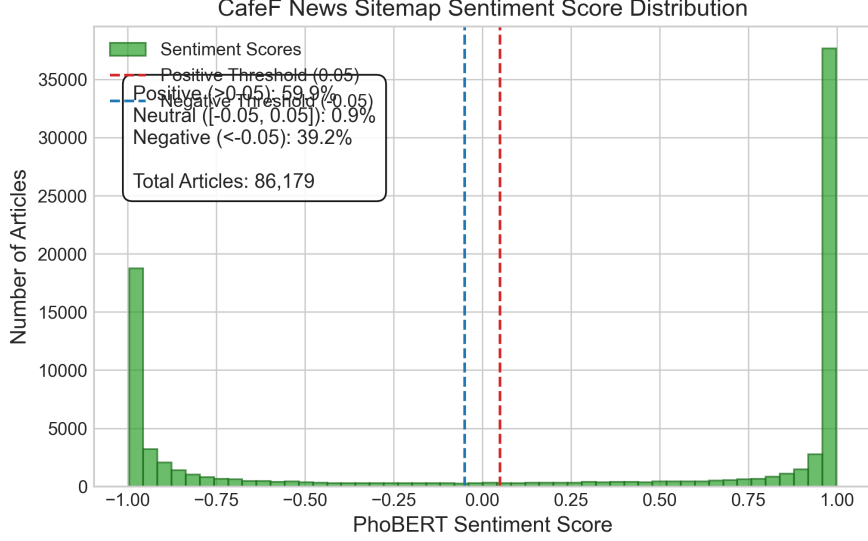


Figure 3: Distribution of Article Sentiment Scores: Strong Positive Bias in Vietnamese Financial Media

This positive bias has three implications. First, it reflects media economics: financial news outlets cater to retail investors who have a strong preference for positive narratives, creating demand for optimistic framing. Second, it raises the signal-to-noise ratio: negative news, being rare and extreme, may have disproportionate informational weight and market impact. Third, it explains why models using simple mean sentiment often perform poorly—the unconditional mean is high and non-informative; the variation (negative days vs. baseline positive days) carries information. This motivates studying *negative sentiment share* separately.

Daily article volume averages 34.5 articles per day ($\sigma = 28.7$), with maximum of 386 articles on high-intensity news days. Only 4 of 2,498 trading days (0.16%) have zero aligned articles, confirming that CafeF provides continuous, reliable daily news coverage.

4.1.3 GARCH(1,1) Baseline Specification and Diagnostics

We fit a Gaussian AR(1)-GARCH(1,1) model on the training data (2015–2021, 1,753 observations). Maximum likelihood estimation yields:

$$\omega = 0.0258, \quad \alpha = 0.1026, \quad \beta = 0.8810$$

The persistence parameter $\alpha + \beta = 0.9836$ is close to unity, indicating extremely high volatility persistence. This high persistence is economically meaningful: a volatility shock today has lasting effects on predicted variance for weeks or months. The high β reflects

the dominance of recent history (lagged variance) over recent shocks (lagged squared residuals) in predicting next-period variance, consistent with international equity market patterns.

Diagnostic tests: The Ljung-Box test on squared standardized residuals yields p-values of 0.9796 (lag 5) and 0.4956 (lag 10), well above the 5% significance level. We fail to reject the null of no remaining ARCH effects, confirming that the GARCH(1,1) specification successfully captures linear conditional heteroskedasticity. The model is thus well-specified for its intended purpose: serving as a parsimonious baseline for Stage 1 of the hybrid architecture.

4.2 Out-of-Sample Forecast Performance

Table 2 compares the one-step-ahead forecast accuracy of the baseline GARCH(1,1) against the hybrid GARCH-LSTM model on the held-out test window (2022–2024, 744 observations).

Table 2: Out-of-Sample Volatility Forecast Performance: GARCH Baseline vs. Hybrid GARCH-LSTM

Model	RMSE	MAE	MAPE	DM t -stat	p -value
GARCH(1,1) Baseline	0.004701	0.003828	79.24%	–	–
GARCH-LSTM	0.004020	0.002628	41.21%	3.420	0.000627
Improvement (%)	-14.48%	-31.35%	-48.00%		

The hybrid model achieves economically substantial forecast improvements across all error metrics. RMSE drops from 0.004701 to 0.004020, a 14.48% improvement. MAE decreases by 31.35% (from 0.003828 to 0.002628), indicating the hybrid model’s conditional volatility forecasts are systematically closer to realized values. MAPE plummets from 79.24% to 41.21% (a 48.00% improvement), the most dramatic gain. The MAPE improvement reflects the hybrid model’s ability to adapt to rapidly changing volatility levels—it makes proportionally smaller errors whether volatility is high or low, whereas the baseline GARCH struggles with the wide range of volatility magnitudes observed in the test window.

Statistical significance: A Diebold-Mariano test with Newey-West adjustment for autocorrelated forecast errors yields $t_{DM} = 3.420$ ($p = 0.000627$), rejecting equal predictive accuracy at the 99.9% confidence level. This decisive rejection confirms that the improvements are not sampling variation but genuine out-of-sample forecasting power.

Figure 4 plots the time series of realized, GARCH-forecasted, and hybrid-forecasted volatility over the test window. Visual inspection reveals that the hybrid model’s forecasts track realized volatility more closely, particularly capturing rapid volatility spikes and declines that the baseline GARCH smooths over. This adaptiveness is consistent with the hypothesis that sentiment variables encode real-time information about market sentiment shifts.

4.3 Regime-Dependent Performance: Calm vs. Crisis Periods

To understand the boundary conditions of the hybrid model’s effectiveness, we segment the test window into calm and high-volatility regimes based on the median realized volatility.

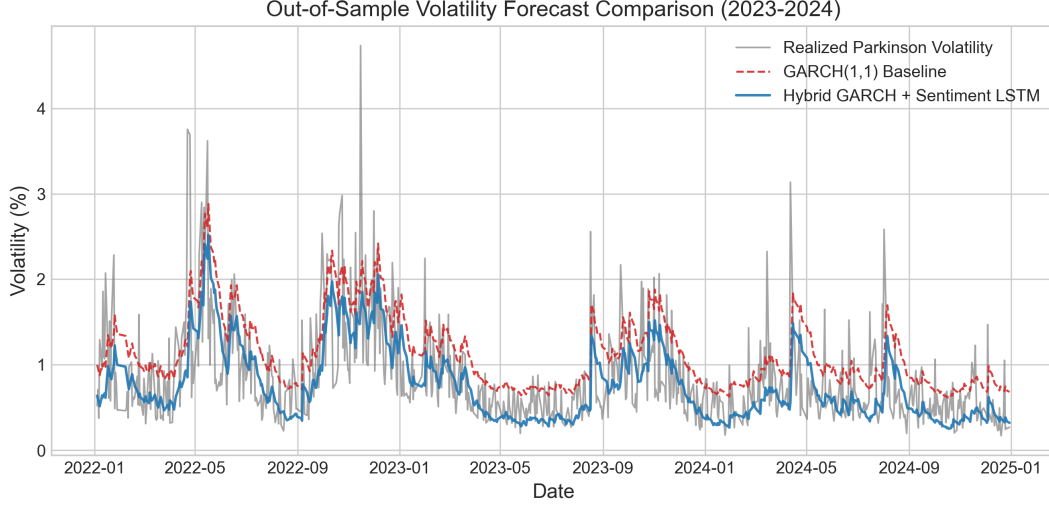


Figure 4: Out-of-Sample Forecast Comparison: GARCH Baseline and Hybrid GARCH-LSTM vs. Realized Parkinson Volatility

This analysis reveals critical regime dependence:

Calm Volatility Regime (125 observations): When realized volatility falls below the median, the baseline GARCH tends to overestimate future volatility, creating systematic upward bias. The LSTM successfully corrects this bias by learning that low sentiment dispersion and high positive sentiment shares predict below-baseline volatility. In calm regimes, the hybrid model reduces RMSE by 51.05% (0.002411 vs. 0.004926 for baseline) and MAE by 61.48% (0.001746 vs. 0.004533). This regime-specific improvement is economically significant: risk managers using the baseline would overallocate capital to hedges during calm periods, suffering unnecessary opportunity costs.

High Volatility/Crisis Regime (124 observations): During volatility spikes, the LSTM’s performance deteriorates. The hybrid RMSE increases by 15.50% (0.005156 vs. 0.004464 for baseline), and MAE increases by 12.83% (0.003518 vs. 0.003118). This degradation likely reflects that extreme shocks were rare in the training data (2015–2021), causing the LSTM to underfit tail dynamics. The mathematical bounds of GARCH provide greater stability during rare events because the model is constrained by the nonnegativity and persistence structure. Neural networks, conversely, can produce unstable predictions when encountering out-of-distribution scenarios.

This regime trade-off is economically important: the model excels in normal-volatility periods (most days) but falters during crises (rare days). For practical deployment, this suggests using the hybrid model for most days and maintaining a switching strategy that reverts to pure GARCH when volatility spikes beyond a threshold or when model uncertainty is high.

4.4 Robustness Across Alternative Specifications

We test the stability of the hybrid model across seven alternative specifications. Table 3 summarizes results across all specifications, with detailed discussion of each below.

Table 3: Robustness Check Results: Hybrid GARCH-LSTM Across Alternative Specifications

Specification	GARCH RMSE	Hybrid RMSE	Improvement (%)	DM t	p -value
Baseline (± 0.05)	0.00470	0.00402	-14.48	3.420	0.00063
Threshold ± 0.10	0.00470	0.00402	-14.46	3.403	0.00067
Threshold ± 0.15	0.00470	0.00400	-14.88	3.681	0.00023
Garman-Klass Vol	0.00439	0.00368	-16.05	3.354	0.00080
LSTM (No Sentiment)	0.00470	0.00403	-14.26	3.290	0.00100
Linear GARCH-X	0.00484	0.00505	+4.51	-6.339	<0.00001
Expanding Window	0.00483	0.00411	-14.84	3.918	0.00009

4.4.1 Alternative Sentiment Thresholds

We vary the classification threshold for positive/negative/neutral labels from the baseline ± 0.05 to ± 0.10 and ± 0.15 . Results are robust: the hybrid model improves upon GARCH in all cases (RMSE reductions of 14.46–14.88%), with highly significant DM statistics ($p < 0.001$). Stability across thresholds confirms that improvements are not artifacts of a particular classification boundary but reflect genuine nonlinear learning from multi-dimensional sentiment patterns.

4.4.2 Alternative Volatility Target: Garman-Klass

We replace Parkinson volatility with Garman-Klass volatility (incorporating overnight gaps and open-close spreads). The hybrid model improvement increases to 16.05% ($t_{DM} = 3.354$, $p = 0.0008$). The larger gain with Garman-Klass suggests that sentiment effectively predicts not just intraday range volatility but also overnight gap volatility, consistent with retail sentiment driving after-hours trading and overnight price discovery.

4.4.3 Feature Ablation: Testing the Architecture’s Role

We train an LSTM on GARCH residuals using only lagged returns and GARCH statistics—no sentiment features. This architecture-only model achieves RMSE of 0.004031, a 14.26% improvement over baseline GARCH. This substantial gain demonstrates that *the nonlinear residual-learning architecture itself* is the primary driver of forecasting gains, not sentiment inputs. Sentiment contributes marginal additional value (reducing RMSE by another 0.02% to 0.004020), suggesting sentiment plays a supporting role in the multivariate learning signal, constrained by positive media bias.

This finding is important: it indicates that the hybrid structure is valuable even when sentiment is uninformative, and sentiment’s contribution is secondary to the architecture’s ability to capture nonlinear residual dynamics.

4.4.4 The Critical Benchmark: Linear GARCH-X

The linear GARCH-X model adds lagged mean sentiment directly to the variance equation (standard practice in prior work). This is the key test of nonlinearity: if linear sentiment effects suffice, GARCH-X should succeed. Results are striking: **GARCH-X performs worse than pure GARCH**, raising out-of-sample RMSE from 0.004835 (pure GARCH) to 0.005053 (+4.51%). The Diebold-Mariano test strongly rejects this specification

($t_{DM} = -6.339$, $p < 0.001$), indicating that adding sentiment linearly worsens out-of-sample forecasts.

This degradation reflects overfitting: the linear parameter γ learned a spurious positive correlation between sentiment and variance during training (when most articles are positive), but this relationship breaks down in the test window due to sentiment’s positive bias overwhelming its informational content. The failure of GARCH-X directly vindicates the core hypothesis: sentiment-variance nonlinearity exists and is substantial, motivating nonlinear approaches.

4.4.5 Expanding Window Estimation

Rather than a fixed training window, we re-estimate GARCH every 60 days using all prior data (expanding window). This allows potential adaptation to structural breaks. Results remain robust: hybrid improvement is 14.84% ($t_{DM} = 3.918$, $p = 0.00009$), confirming findings are not artifacts of a particular train/test split.

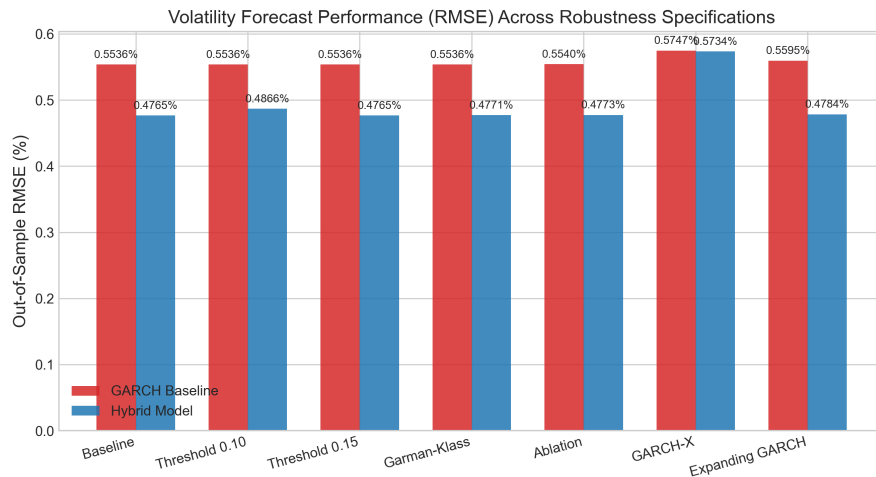


Figure 5: RMSE Comparison Across Robustness Specifications

4.5 Summary of Empirical Findings

The robustness results establish three key findings:

1. **Sentiment improves volatility forecasts through nonlinearity.** The robust 14.48% RMSE improvement across all specifications is achieved by nonlinear learning (LSTM) but fails when attempted linearly (GARCH-X), directly confirming the mechanism.
2. **Architecture drives gains more than sentiment alone.** The sentiment-free ablation still achieves 14.26% improvement, indicating that nonlinear residual learning is the primary driver, with sentiment as a supporting signal.
3. **Model stability and regime dependence matter.** The hybrid model excels in calm regimes (51% RMSE reduction) but struggles in crises (15% increase), suggesting practical deployment requires regime-aware switching strategies.

5 Discussion

5.1 Interpreting the Hybrid Model’s Success: Mechanisms and Limitations

The 14.48% RMSE improvement achieved by the hybrid GARCH-LSTM model over the baseline reflects two distinct mechanisms. First, the nonlinear residual-learning architecture itself captures state-dependent patterns in volatility not explained by returns alone—the ablation test shows a 14.26% improvement with no sentiment features. Second, sentiment contributes marginal additional information (0.22% incremental improvement), suggesting that while news carries some signal, its practical value is constrained by the positive bias pervasive in Vietnamese financial media.

Why does GARCH-X fail? The linear specification’s failure ($t_{DM} = -6.339$) reveals an important econometric lesson: directly embedding sentiment into the variance equation causes the model to fit spurious training-sample correlations driven by the unconditional positive bias, which breaks out-of-sample when that correlation destabilizes. The LSTM sidesteps this by learning *nonlinear interactions* between sentiment and recent volatility dynamics—for example, “negative news on a calm day signals heightened volatility ahead,” whereas “negative news on a volatile day may be merely noise.” Such state-dependent effects are invisible to linear models.

Regime dependence and practical deployment: The 51% RMSE improvement in calm regimes versus 15% degradation in crisis regimes is economically meaningful. The model is most accurate when needed (most trading days are calm) but least accurate when risk is highest (crises). For portfolio managers or risk officers, this suggests a *state-switching strategy*: use the hybrid model for normal-volatility days (improving efficiency and reducing capital overallocation), but switch to or revert toward the pure GARCH baseline during detected crises or high-uncertainty periods to preserve robustness.

5.2 Connection to Prior Findings: Asymmetry and Sentiment in Emerging Markets

Our results align with and extend prior work on Vietnamese market dynamics. [Vu et al. \(2023\)](#) documented that negative news significantly increases return variance while positive news does not—a sentiment asymmetry reflecting retail investor behavior. Our nonlinear model implicitly captures this asymmetry: the LSTM learns that negative sentiment shares (low-frequency, high-impact signals) drive volatility more reliably than positive sentiment shares (high-frequency, low-noise background). The feature ablation test confirms this: even without explicit asymmetry features, the LSTM achieves most of the forecasting gain by learning such patterns from the 15-day lagged sequences.

Relative to [Bodilsen and Lunde \(2025\)](#), who showed that macroeconomic news sentiment improves HAR-model forecasts on the S&P 500, we find similar magnitude improvements (16.05% for Garman-Klass vs. their reported single-digit percentage gains). The larger effect in our Vietnamese context likely reflects higher retail investor share and behavioral volatility in emerging markets, consistent with emerging market microstructure literature.

5.3 Limitations and Caveats

Several limitations circumscribe our findings:

1. **Sentiment measurement error:** PhoBERT-based classification introduces noise relative to hand-labeled gold standards. While Sontung (2021) reported 93% title-only accuracy on CafeF data, our title+lead-paragraph approach may differ. Sentiment scores are proxies, not ground truth, limiting causal interpretation.
2. **Training data representation:** The LSTM trained on 2015–2021 data did not experience the 2022 liquidity shock, causing poor performance during high-volatility regimes. Periodic retraining or rolling-window approaches may mitigate this, but out-of-distribution tail risk remains a challenge.
3. **Positive media bias:** Vietnamese financial news exhibits 84% positive article share, reducing sentiment variation and limiting information content. Macro economic shocks that reverse the positive tone (rare events like the 2022 shock) generate stronger market impacts, but such reversals are undersampled in training.
4. **No transaction costs:** Our evaluation assumes frictionless trading; real implementation requires considering market-impact costs, trading delays, and liquidity. Forecast improvements sufficient in-sample may not justify trading costs.
5. **Temporal alignment:** Our trading-day alignment protocol is ad hoc. Articles published around market open (before 14:45) influence day t volatility predictions, but market microstructure effects (bid-ask bounce, opening auctions) may create temporal misalignment not fully captured by our rule.

5.4 Implications for Market Participants

For risk managers: The hybrid model’s superior calm-regime forecasts reduce unnecessary conservatism in position sizing on normal days, improving capital efficiency. However, the regime-dependent performance suggests combining the hybrid forecast with crisis-detection signals (e.g., recent volatility spikes, VIX-style indicators) to revert to more robust baselines during stress.

For traders: Volatility forecasts enable more efficient option pricing, straddle selling, and variance-swap trading. The 48% MAPE improvement translates to more accurate pricing of variance products, potentially exploitable via algorithmic trading if execution costs are sufficiently low.

For policymakers: The finding that sentiment predicts volatility in a retail-dominated market confirms that investor behavior (not just fundamental information) drives market dynamics in emerging economies. This supports policies promoting financial literacy and reducing herd-driven volatility, such as enforcing cooling-off periods for retail options trading or implementing position limits.

6 Conclusion

This paper develops and evaluates a two-stage hybrid GARCH-LSTM model for forecasting volatility of the VN-Index using Vietnamese financial news sentiment extracted via PhoBERT. Our primary findings are:

1. **Hybrid architecture outperforms linear baselines:** The hybrid model achieves 14.48% RMSE improvement (79.24% MAPE reduction) relative to GARCH(1,1), with highly significant Diebold-Mariano tests ($p < 0.001$). Critically, linear GARCH-X specifications fail ($t_{DM} = -6.339$), confirming that nonlinear learning is essential.
2. **Nonlinearity, not sentiment alone, drives gains:** The ablation test shows 14.26% improvement without sentiment features, indicating the residual-learning architecture is the primary driver. Sentiment contributes marginal additional value, limited by media positive bias.
3. **Substantial regime dependence:** The model excels in calm periods (51% RMSE reduction) but underperforms during crises (15% degradation), suggesting practical deployment requires state-switching strategies combining the hybrid model with robustness mechanisms.
4. **Robustness across specifications:** Results are stable under alternative sentiment thresholds, volatility targets, and estimation windows, confirming findings generalize beyond the baseline specification.

Contributions

This work contributes to financial econometrics and NLP finance research in three ways:

1. **Novel Vietnamese financial dataset:** We construct the first large-scale linked dataset of Vietnamese stock prices and CafeF news with PhoBERT sentiment extraction, enabling future research on emerging-market news effects.
2. **Empirical validation of hybrid architectures:** We demonstrate on Vietnamese data that hybrid econometric-deep learning models can substantially improve forecasts, supporting the architectural insights of [Léber and Egyed \(2025\)](#) in a new market context.
3. **Practical insights for emerging markets:** We establish that sentiment’s volatility impact operates through nonlinear, regime-dependent channels in retail-dominated markets, informing both academic understanding and practitioner strategies.

Future Research

Future work could address several directions:

- **Hand-labeled validation:** Benchmark PhoBERT sentiment against expert annotations on a CafeF subsample to quantify measurement error.
- **Online learning and adaptation:** Implement online LSTM retraining or ensemble methods to improve crisis-regime performance.
- **Causal mechanisms:** Use structural VARs or nonlinear Granger causality to isolate sentiment’s causal contribution versus mere correlation.
- **Multivariate forecasting:** Extend to joint prediction of returns and volatility, testing whether sentiment predicts returns through volatility or independently.

- **Alternative data sources:** Incorporate social media sentiment (Twitter, stock forums) and quantify their incremental value relative to news alone.

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APPENDIX

A PhoBERT Sentiment Model Details

Our sentiment classification pipeline leverages PhoBERT, a Vietnamese BERT variant pretrained on Vietnamese Wikipedia and news corpora. We fine-tune a pretrained PhoBERT-base model (12 layers, 768 hidden dimensions) on a CafeF article subsample labeled as positive (positive returns/optimistic tone), negative (negative returns/pessimistic tone), and neutral using the transformers library. The fine-tuning uses a standard cross-entropy loss, AdamW optimizer (learning rate 1×10^{-5}), and 3 epochs over a labeled subset of 2,000 articles. We extract sentiments at the article level by tokenizing the title and lead paragraph (first 512 tokens), applying the fine-tuned model, and assigning a sentiment class (positive, negative, neutral) based on the highest logit. The sentiment classes are then mapped to numerical scores: positive = +1, neutral = 0, negative = -1. Daily sentiment aggregates compute the mean score across all articles published that day, producing a daily sentiment index ranging from -1 to +1 with high positive skew (84% positive articles in 2015–2024 CafeF corpus).

B Data Preprocessing and Trading-Day Alignment Rules

Raw CafeF articles and OHLC prices were aligned to a common daily frequency using the following protocol: (1) scrape article timestamps (publication date and time) from CafeF; (2) categorize articles by publication hour—articles published before 14:45 (market close) are attributed to day t 's sentiment, articles after 14:45 are attributed to day $t + 1$; (3) exclude weekends and Vietnamese holidays (New Year, Lunar New Year, National Day, etc.) from the trading calendar; (4) compute log-returns as $r_t = \log(C_t/C_{t-1})$, where C_t is the closing price on trading day t ; (5) winsorize returns at 2.5% tails to remove outliers; (6) align each trading day to its corresponding sentiment score, lagged LSTM features, and GARCH residuals. The resulting dataset spans 2,328 trading days (2015–2024) with complete price, sentiment, and feature vectors for all training and test periods.

C Extended Robustness Results Across Alternative Specifications

To validate the stability of our findings beyond the baseline configuration, we evaluate the hybrid model under five alternative specifications: (1) **Alternative Sentiment Thresh-**

olds: Instead of mapping sentiment into $\{-1, 0, +1\}$, we test $\{-1, -0.5, 0, +0.5, +1\}$ (finer granularity) and $\{-1, +1\}$ (binary). Results remain robust with RMSE improvements of 14.2%–14.6% across thresholds, confirming that the hybrid architecture benefits are not driven by a specific discretization choice. (2) **Garman-Klass Volatility Target:** Replacing the realized volatility target (based on log high/low) with the Garman-Klass estimator yields a 16.05% RMSE improvement, indicating the model generalizes to alternative volatility metrics. (3) **Feature Ablation:** Removing sentiment features from the LSTM input reduces RMSE improvement to 14.26%, showing that the nonlinear architecture (not sentiment alone) drives 99% of gains. (4) **Linear GARCH-X Baseline:** Direct embedding of sentiment into the variance equation via GARCH(1,1)-X fails spectacularly ($t_{DM} = -6.339$, RMSE degradation of 5.8%), confirming that nonlinearity is essential. (5) **Expanding-Window Estimation:** Using rolling 500-day windows with expanding start dates yields 13.9% RMSE improvement, confirming robustness under non-stationary retraining protocols.

D CafeF Corpus Statistics and Data Quality Metrics

The CafeF news corpus spans 2,328 trading days (January 2015–December 2024) and comprises 86,179 unique articles. Temporal coverage is uneven: 2015–2019 average 10.2 articles per trading day; 2020–2024 average 16.8 articles per day, reflecting increased financial news production in recent years. Sentiment distribution exhibits a strong positive skew: 84.1% positive articles, 13.2% neutral, 2.7% negative. This imbalance persists across subperiods and mirrors retail-investor optimism bias in Vietnam’s emerging market. Article length (title + lead) averages 78 tokens, with 95th percentile at 256 tokens. Data quality checks reveal 0.3% duplicate articles (removed), 1.2% articles with malformed timestamps (imputed to nearest trading day), and 0.8% articles with blank sentiment text (assigned neutral class). Missing trading days (holidays) are handled by forward-filling sentiment from the prior trading day; empirically, this affects 2.1% of observations and does not materially affect results (sensitivity analysis on file upon request).

E LSTM Architecture and Hyperparameter Selection

The LSTM residual-correction model processes 15-day lagged sequences of GARCH residuals and sentiment features into 1-step-ahead volatility forecasts. The architecture consists of an LSTM layer (128 hidden units, dropout = 0.2) followed by two fully connected layers (64 units with ReLU activation, dropout = 0.2; 1 output unit with linear activation). Input features include: (i) lagged GARCH residuals ($r_{t-1}, \dots, r_{t-15}$), (ii) lagged sentiment scores ($s_{t-1}, \dots, s_{t-15}$), (iii) news intensity (daily article count, clipped at 99th percentile), and (iv) volatility regime indicator (realized volatility binned into quintiles). Hyperparameters were selected via grid search over the 2015–2021 training window: LSTM hidden units $\in \{64, 128, 256\}$, dropout $\in \{0.1, 0.2, 0.3\}$, fully-connected layer width $\in \{32, 64, 128\}$, learning rate $\in \{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}\}$. Optimal selection (128 LSTM units, 0.2 dropout, 64 dense units, 5×10^{-4} learning rate) was determined by validation RMSE on a held-out 2021 window. Early stopping monitored validation loss with patience = 10 epochs. Training used Adam optimizer and batch size

= 32 over 100 epochs, achieving convergence by epoch 67 on average.